Experiment No. 7

Post-Lab Questions – CrewAI Multi-Agent System

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Q1. Analyze how task coordination and collaboration are achieved among the Research Agent, Content Writer Agent, and Reviewer Agent in the implemented CrewAI system.

Answer: Task coordination and collaboration are achieved by dividing the complete report-generation process into separate tasks and assigning each task to a specialized agent. The **Research Agent** is responsible for collecting, analyzing, and organizing relevant information about the given research topic. After completing the research task, its output is passed as context to the **Content Writer Agent**. The Content Writer uses this research information to prepare a structured and meaningful report. The generated report is then passed to the **Reviewer Agent**, which checks the report for accuracy, completeness, clarity, consistency, and proper organization. CrewAI manages this information flow using task dependencies and sequential execution. Therefore, the agents do not work independently; instead, the output of one agent becomes the input or context for the next agent. This coordination creates a pipeline such as **Research** → **Writing** → **Review**, making the overall workflow organized and easier to manage.

Q2. Compare the execution of the CrewAI workflow with and without task delegation. Discuss its impact on agent interactions and workflow efficiency.

Answer: Without task delegation, every agent performs only its predefined responsibility and follows the fixed workflow of **Research Agent** → **Content Writer Agent** → **Reviewer Agent**. This approach is simple, predictable, and generally faster because there are fewer agent-to-agent interactions and fewer LLM calls. However, an agent cannot directly request additional assistance when it faces a task that requires information or expertise outside its assigned responsibility. With task delegation enabled, an agent can request assistance from another suitable agent when required. This creates more interaction and collaboration between agents and can be useful for complex tasks that require additional research, verification, or specialized knowledge. The main advantage of delegation is improved flexibility and potentially better results. However, delegation may also increase the number of LLM calls, processing time, and computational or API usage. Hence, a workflow without delegation is more efficient for simple and well-defined tasks, while delegation is more useful when the workflow requires dynamic collaboration between agents.

Q3. Suppose the Research Agent generates incomplete or inaccurate information. Propose a suitable modification to the existing CrewAI workflow to improve the reliability of the final report.

Answer: If the Research Agent produces incomplete or inaccurate information, a suitable improvement would be to introduce a separate **Research Verification Agent** between the Research Agent and the

Content Writer Agent. The Research Agent would first collect the required information, and the Verification Agent would then examine the research for missing details, inconsistencies, unsupported claims, and possible factual errors. If the information is found to be incomplete or incorrect, the research can be revised or additional information can be collected before it is provided to the Content Writer. The improved workflow would therefore be **Research Agent** → **Research Verification Agent** → **Content Writer Agent** → **Reviewer Agent**. The final Reviewer Agent would still perform the last quality check on the complete report. This additional verification stage provides an extra quality-control checkpoint and reduces the possibility of incorrect research being directly converted into the final report. It therefore improves the reliability, accuracy, and overall quality of the generated output.

Conclusion: The CrewAI system demonstrates effective multi-agent collaboration by dividing the report-generation process among specialized agents and passing outputs between dependent tasks. Sequential execution provides a controlled workflow, while task delegation allows greater flexibility and interaction when assistance is required. Adding a research verification stage can further improve the reliability and accuracy of the final report.